M2-Mixer: A Multimodal Mixer with Multi-head Loss for Classification from Multimodal Data

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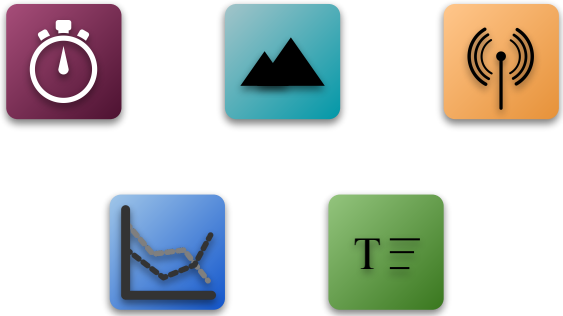
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Outline

1. Context
2. Related Work
3. Proposed Approach
4. Experiments & Results
5. Conclusion & Future work

Why We Need Multimodality?

Modalities

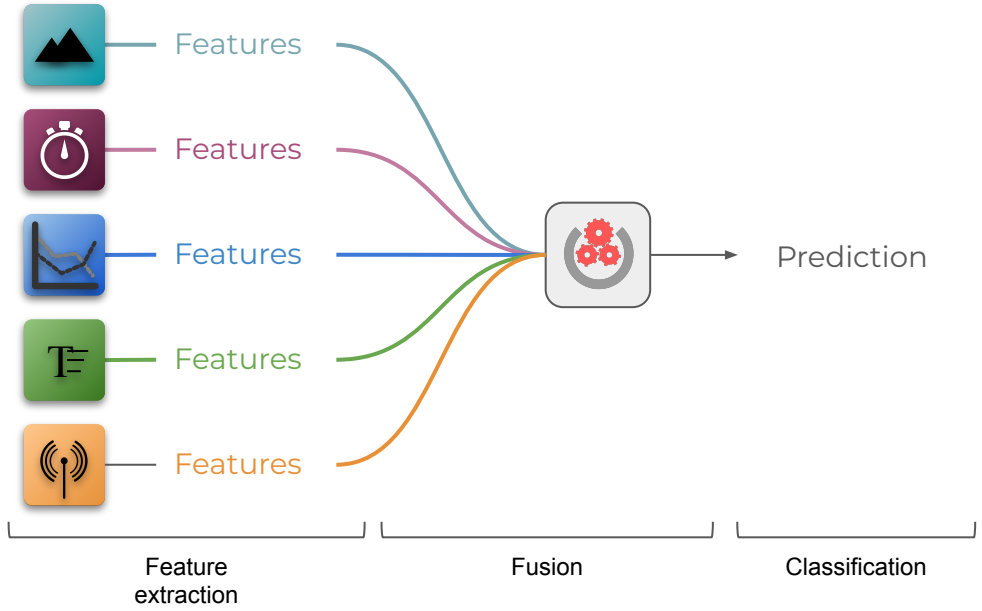


Application
Use cases



Multimodal Deep Learning allows models to learn more comprehensive and accurate representation of the real world

What is Multimodal Classification?



Problem: Fuse modalities effectively to achieve high performance with as little time complexity as possible

Related Work - Multimodal Classification

Current state-of-the-art model mainly use

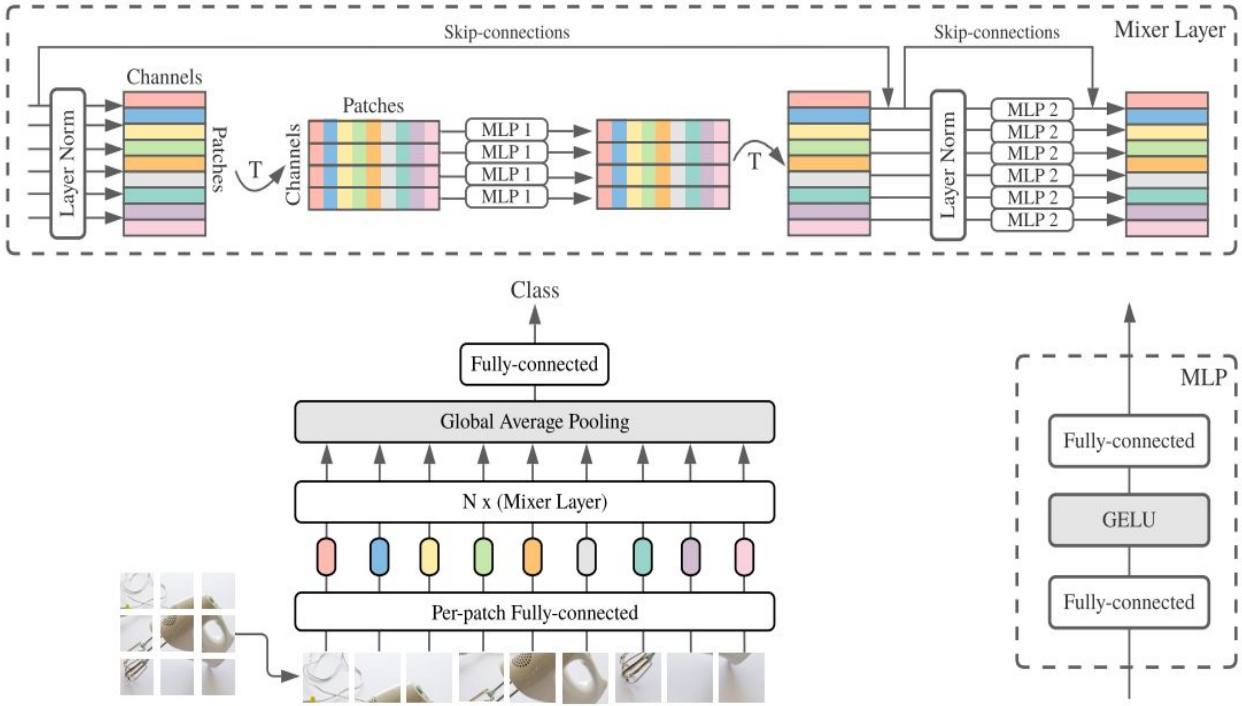
- Big Convolutional Networks
- Transformers
- Neural Architecture Search
- Pre-trained models
- Complex fusion functions

These approaches are often conceptually, computationally complex

Multimodal Networks may favor one modality over the other, and find suboptimal representations for the modalities [Wang et al., 2020]

Related Work - MLP-Mixers

- Only MLPs
- Conceptually / Computationally Simple
- Competitive with Transformers / CNNs



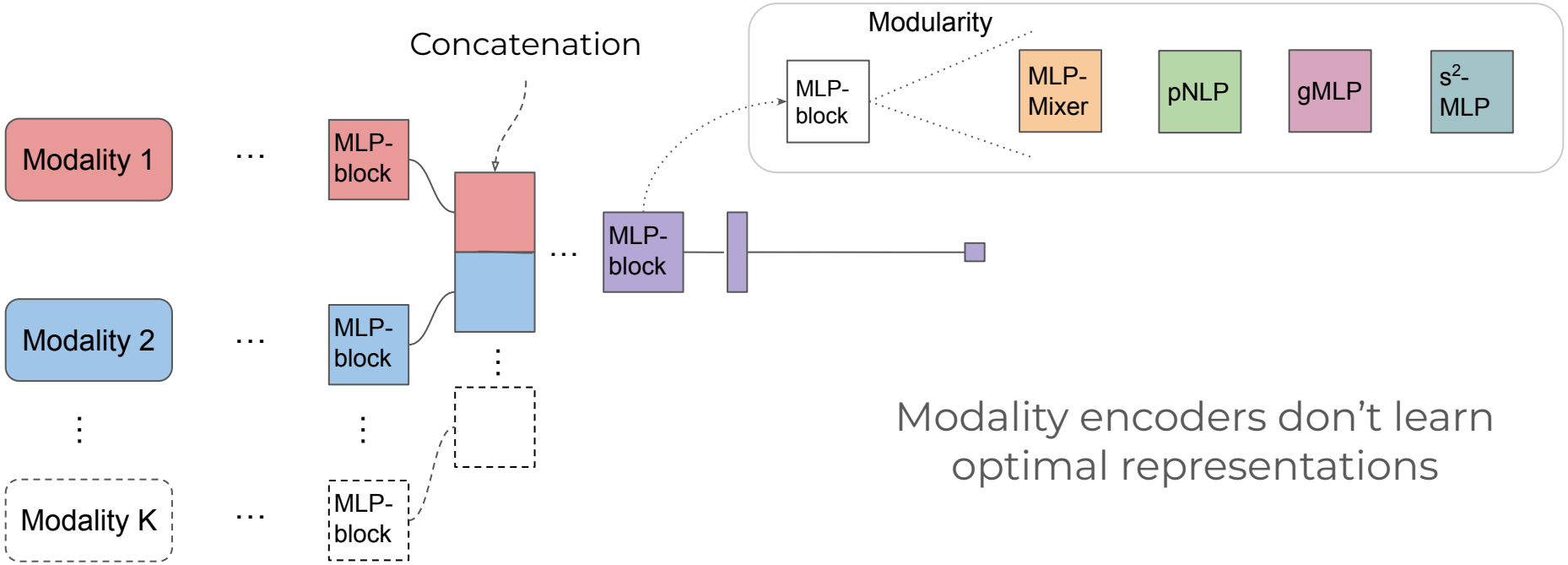
(Tolstikhin et. al, 2021)

Main Contributions

Contribution 1:
Propose an all MLP-based approach for multimodal fusion

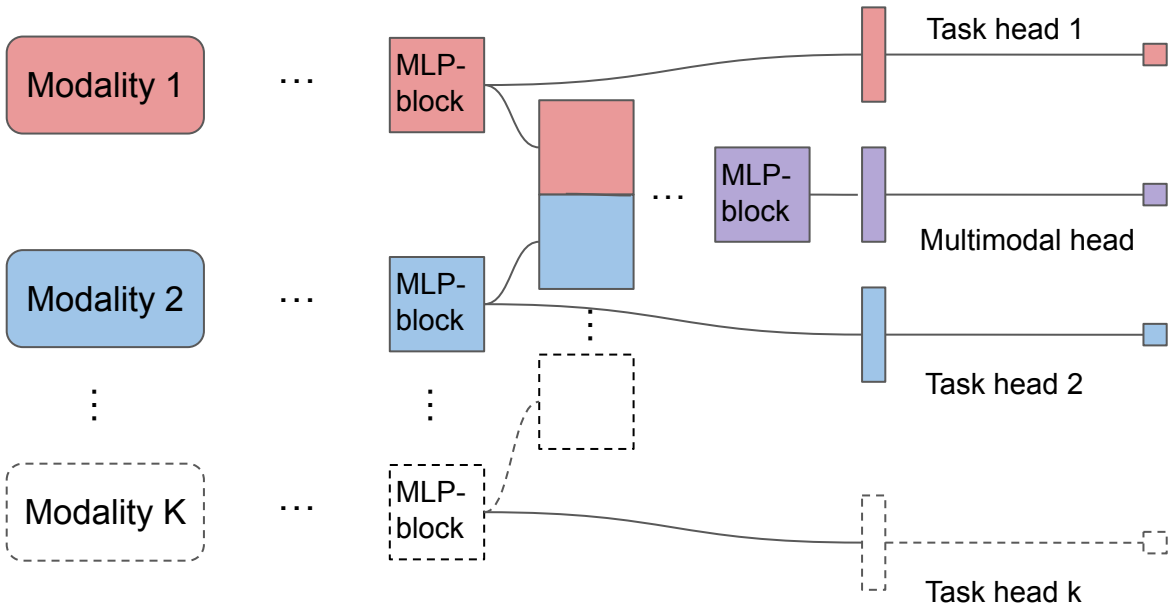
Contribution 2:
Improve modality representations by optimizing the learning process with multi-head loss

Multimodal Mixer: Use MLP-Blocks for Feature Extraction and Fusion

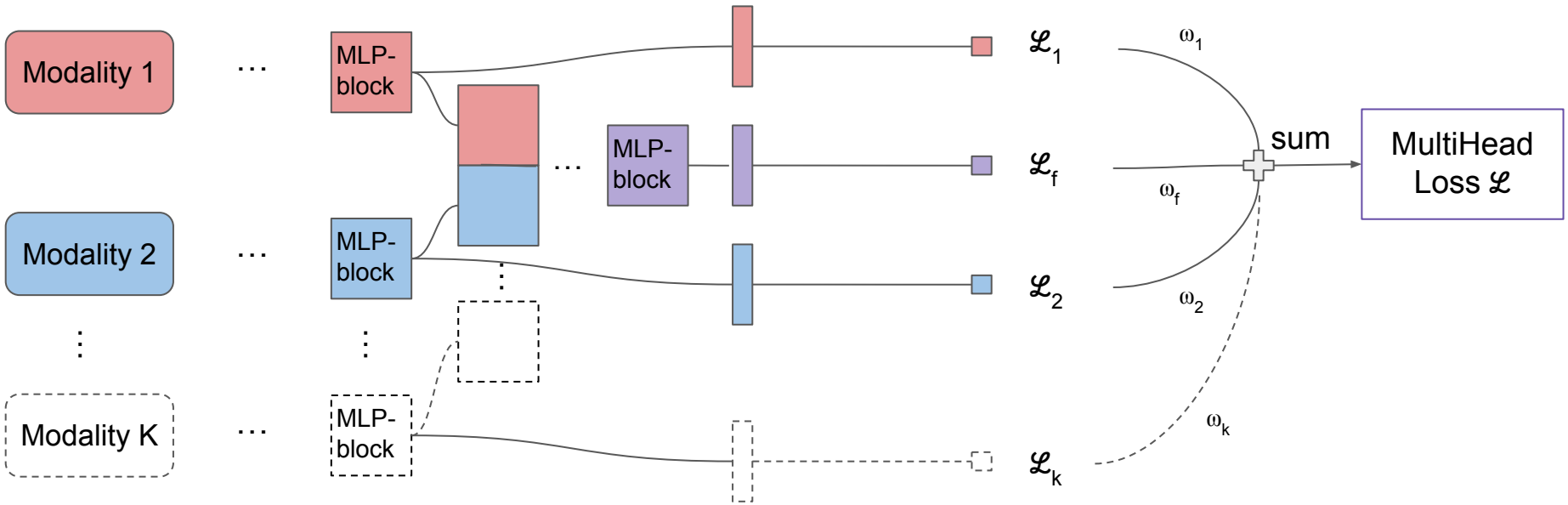


Modality encoders don't learn optimal representations

M2-Mixer: Add Task Heads to Modality Encoders



M2-Mixer: Optimize Using Weighted Loss



$$\mathcal{L}(\hat{y}_f, \hat{y}_1, \hat{y}_2, y) = w_f \mathcal{L}_f(\hat{y}_f, y) + w_1 \mathcal{L}_1(\hat{y}_1, y) + w_2 \mathcal{L}_2(\hat{y}_2, y) + \cdots + w_k \mathcal{L}_k(\hat{y}_k, y)$$

Experiments

2 Datasets:

	AV-MNIST [Vielzeuf et al., 2018]	MIMIC-III [Johnson et al., 2015]
Field	Multimedia	Healthcare
Modalities	Image / Audio	Time Series / Tabular
Samples train / val / test	55,000 / 5000 / 10000	26,093 / 3,261 / 3,261

Our models:

- MMixer (no multi-head loss)
- M2-Mixer (with multi-head loss)

9 Baseline models:

Simple Late Fusion [Liang et al, 2021], LRTF [Liu et al., 2018], MFAS [Pérez-Rúa et al., 2019], RefNet [Sankaran et al., 2021], MVAE [Wu et al., 2018], MFM [Tsai et al., 2019], CCA [Sun et al., 2020], MI-Matrix [Jayakumar et al., 2020], GradBlend [Wang et al., 2020]

Results of Multimodal Mixer on AV-MNIST

No multi-head loss

Results after 10 runs

Results on **AV-MNIST** dataset.

Architecture	Acc. (%) (avg)	Acc. (%) (max)	F-1 (%) (avg)	Training Time (s)	Test Time (s)	Training Params (M)
Simple Late Fusion	71.55 ± 0.6	72.30	70.91 ± 0.7	873 ± 5	5.5 ± 0.1	0.26
LRTF	71.53 ± 0.4	72.30	70.98 ± 0.4	1108 ± 30	5 ± 0.2	0.25
MFAS	72.63 ± 0.2	72.92	72.08 ± 0.2	65710 ± 12817	12.3 ± 1	0.14 [2]
RefNet	71.60 ± 0.8	72.75	70.79 ± 0.9	891 ± 5	7.5 ± 5	14.1
MVAE	70.72 ± 1.2	72.43	69.99 ± 1.0	1129 ± 12	5.6 ± 0.2	0.81
MFM	71.40 ± 0.8	72.28	70.76 ± 0.9	1110 ± 5	5.4 ± 0.13	0.92
CCA	72.01 ± 0.4	72.76	71.42 ± 0.5	$\mathbf{378 \pm 1}$	$\mathbf{3.48 \pm 0.07}$	0.25
MI-Matrix	71.14 ± 0.4	71.83	70.45 ± 0.5	631 ± 2	5.5 ± 0.2	2.53
GradBlend	68.71 ± 0.7	69.51	68.52 ± 1.3	43768 ± 5554	8.1 ± 2	0.29
MMixer	71.58 ± 0.6	72.66	71.10 ± 0.6	1632 ± 158	5.3 ± 0.25	0.88

MMixer achieves comparable results to the other models

Analyzing Multimodal Mixer on AV-MNIST

No multi-head loss

Accuracies (%) of feature extractors of different models on AV-MNIST dataset

Modality	Uni-modal models	MMixer
Audio	39.93	35.58
Image	66.10	63.95

Feature encoders learn sub-optimal representations

Adding Multi-head Loss

Accuracies (%) of feature extractors of different models on AV-MNIST dataset

With multi-head loss with uniform weights

Modality	Uni-modal models	MMixer	M2-Mixer_M
Audio	39.93	35.58	39.90
Image	66.10	63.95	66.10

Multi-head loss forces modalities to learn close-to-optimal representation

Results of M2-Mixer

AV-MNIST

Architecture	Accuracy % - avg (10 runs)	Accuracy % - max	Train time (s)
MFAS	72.64 ± 0.2	72.93	65710 ± 12817
GradBlend	68.71 ± 0.7	69.51	43768 ± 5554
M2-Mixer B	**73.06 ± 0.2**	**73.34**	10271 ± 6578
M2-Mixer M	72.81 ± 0.2	73.20	**4147 ± 1642**

MIMIC-III

Architecture	Accuracy % - avg (10 runs)	Accuracy % - max	Train time (s)
MFAS	78.02 ± 0.4	78.63	8043 ± 663
GradBlend	78.1 ± 0.3	78.51	7988 ± 239
M2-Mixer H	78.32 ± 0.3	**79.03**	840 ± 119
M2-Mixer LC	**78.43 ± 0.3**	78.76	**597 ± 113**

Our Proposed Models

B, M, H, LC are different configurations of the same model

Number of Parameters

B : 8.3 m
M : 88 k
H : 2.9 k
LC : 2.9K

M2-Mixer outperforms MFAS and GradBlend with much lower training time

Conclusion & Future Work

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- Proposed MMixer: an all-MLP based architecture for multimodal fusion
- Proposed M2-Mixer: MMixer with Multi-head Loss to Improve modality representations
- Test M2-Mixer with additional datasets
- Implement better loss weighting approach
- Test other MLP-based architectures for feature extraction and fusion

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